

Compute Commons: Usable Access to Robot Inference

Xinrong Sun FP9 · Compute Commons Duke Kunshan University

Course instructor: Professor Luyao Zhang



RESEARCH REVISION · 5 OCTOBER 2026

KEY MESSAGE

Robot state changes which compute allocations are usable.
Physical feasibility also changes the incentive to bid truthfully.

01 Question & model

Research question & verified gap

Can access policy deliver usable robot inference while preserving truthful value bids?
RTC studies delayed actions [1]. Deadline-aware planning already allocates effort [2]. Our comparison connects usable access to payments and entry priority.

Strategic model & foundations

Four operators: values A = 10, B = 9, D = 8, E = 6. One server, 400 ms, 200 ms per inference. Only E is an eligible entrant. Deadline = min(queue time, pose-validity time) minus network delay. State is public; value is private.

Three complementary lenses

Game theory: truthful values require timely service and critical payments.
Social choice: usable entrant access can cost task value.
Mechanism design: optimize credited bids over physically feasible schedules.

Auction or allocation comparison

Enumerate feasible subsets; dispatch in deadline order. Charge each winner the bid threshold needed to enter a winning set. Losers pay zero.
One sealed round ends after execution and settlement. Telemetry and eligibility must be verified.

02 Design & evidence

Robot-state constraints define usable service		
Robot	Queue / pose time	Deadline
A	200 / 400 ms	200 ms
B	400 / 200 ms	200 ms
D	400 / 400 ms	400 ms
E	400 / 400 ms	400 ms

Figure 1. A has only 200 ms of queued actions. B has 400 ms of actions, but target motion makes its pose stale after 200 ms. All numbers are synthetic.

Actual synthetic comparison: usable value					
Rule	Admit	Pay	Usable	Value	Rev.
Generic	A, B	8 / 8	A	10	16
EDF	A, D	0 / 0	A, D	18	0
Feasible	A, D	9 / 6	A, D	18	15
Credit 3	A, E	9 / 5	A, E	16	14

Generic admission gives B an expired result and utility −8. B can gain by bidding zero.
Feasible admission restores the stated incentive domain. Credit 3 exchanges 2 value units for usable access by E.
36,289 checks; 300 paired synthetic batches. Run code/robot_model.py.

03 Impact & dialogue

Proposed case: shared manipulation inference

Small integrators and incumbent operators share a tabletop pick-and-place inference service.
Measure queue depth, object motion, latency and accepted actions. Workers control reserved evaluation time. Operators can appeal state or eligibility errors. Local controllers handle protective stopping.

Interdisciplinary evidence

Behavior: predict a payment and timely completion before seeing the result; export reflection and regret. No human findings are claimed.
Computation: exact feasible allocation, EDF baseline, state ablations and network-delay stress.
In 300 synthetic batches, mean usable value is 43.06 for feasible allocation and 38.86 for EDF. These are model outcomes, not robot measurements.

Limitations & next test

No robot trials, learned policy, safety proof or dynamic equilibrium. Exact enumeration is an offline oracle.
Next: calibrated traces, solver-overhead measurement, controlled robot trials and a comprehension pilot.

Reserve 100 ms: commercial value 10.
Reserve + credit 5: E wins, value 6.

Acknowledgments

Xinrong Sun developed the original project and identified the revision need. Thanks to Professor Luyao Zhang for feedback and templates. Codex assisted revision; disclosure is in the paper.

Selected references

[1] Black et al. (2025). Real-Time Execution of Action Chunking Flow Policies. arXiv:2506.07339
[2] Sung et al. (2024). Deadline-Aware Task and Motion Planning. arXiv:2410.05828
[3] Mahajan et al. (2020). Themis. NSDI. [4] Roughgarden (2013). Myerson’s Lemma.

OPEN MATERIALS

GitHub: dku-comsci-econ206-Autumn2026 / -PS2-FP9-ComputeCommons
HF: dku-comsci-econ206-2026 / compute-commons
Source and release status accompany the package.